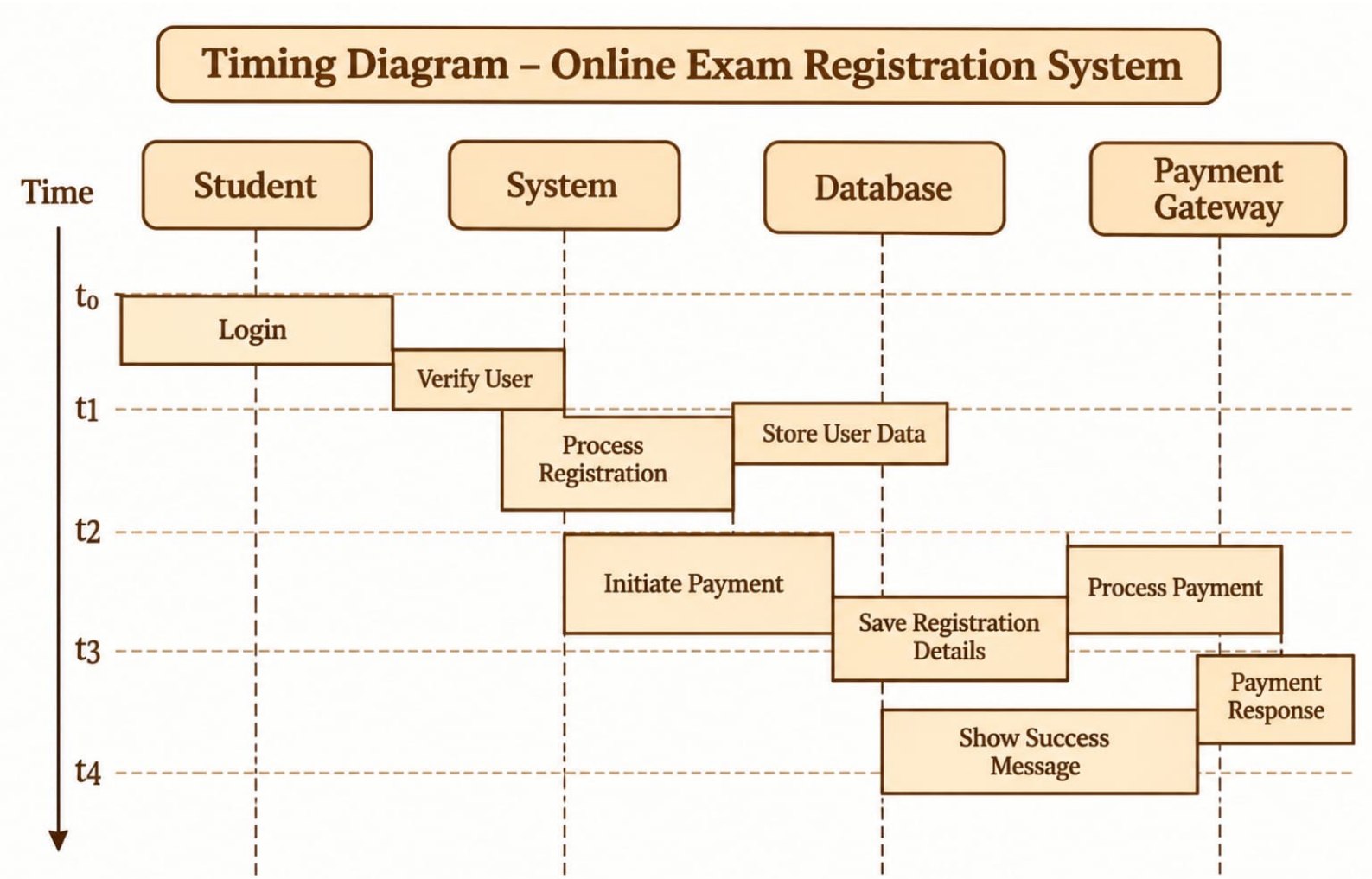
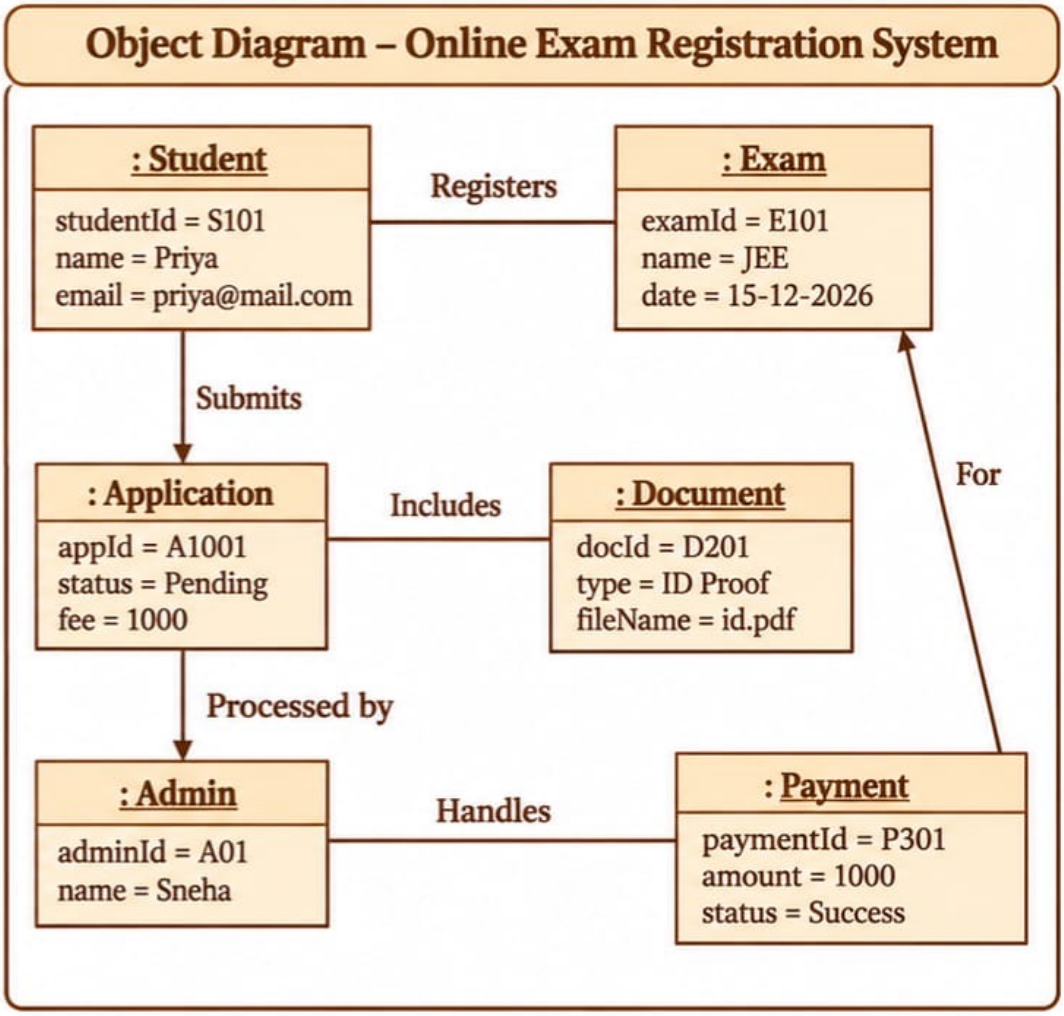
Problem Statement:

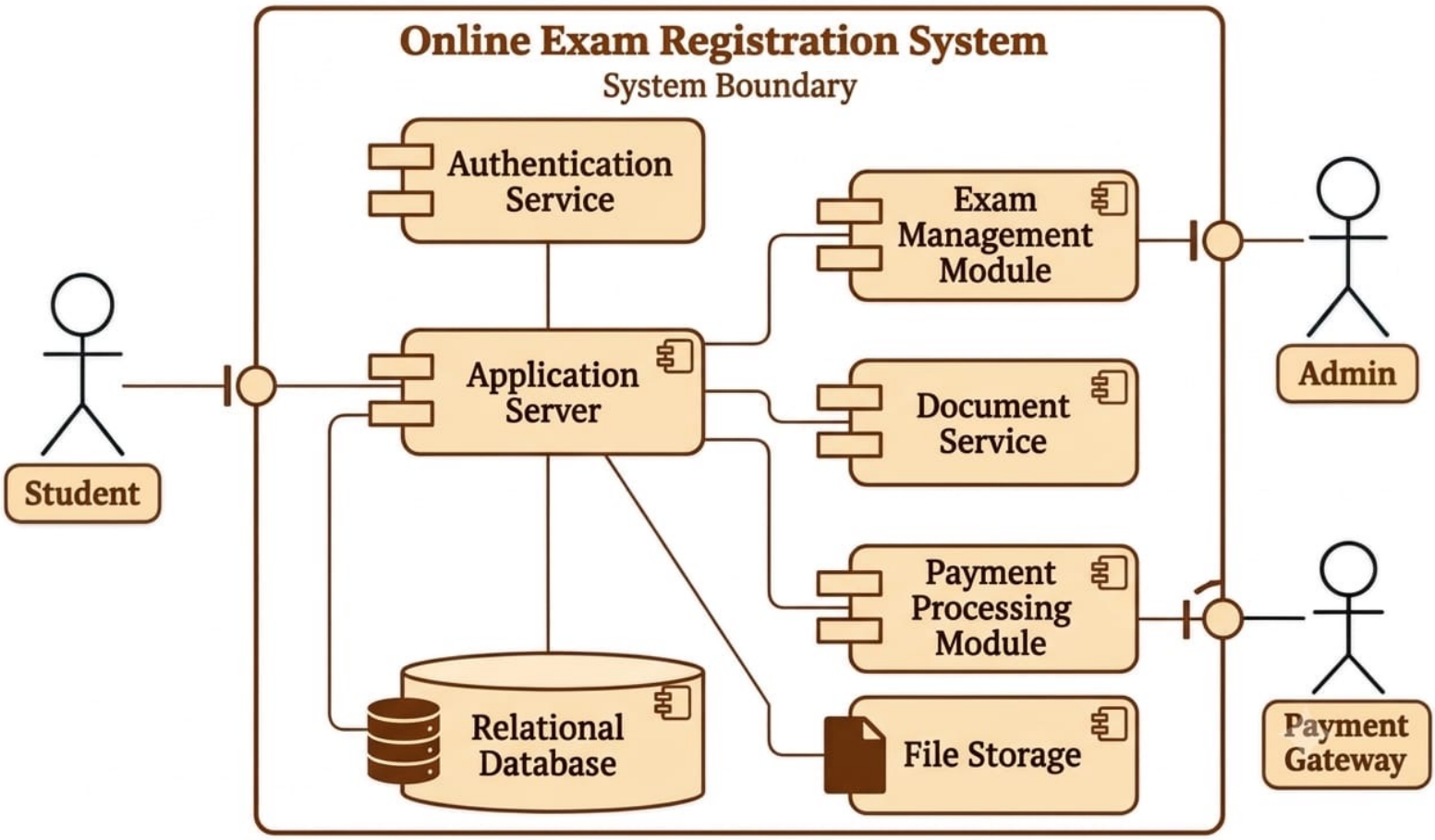
The current exam registration process is manual, time-consuming, and prone to errors such as duplicate entries and incorrect student details. Students must visit the institution to submit forms and documents, while staff face difficulty in verifying and managing large numbers of applications.

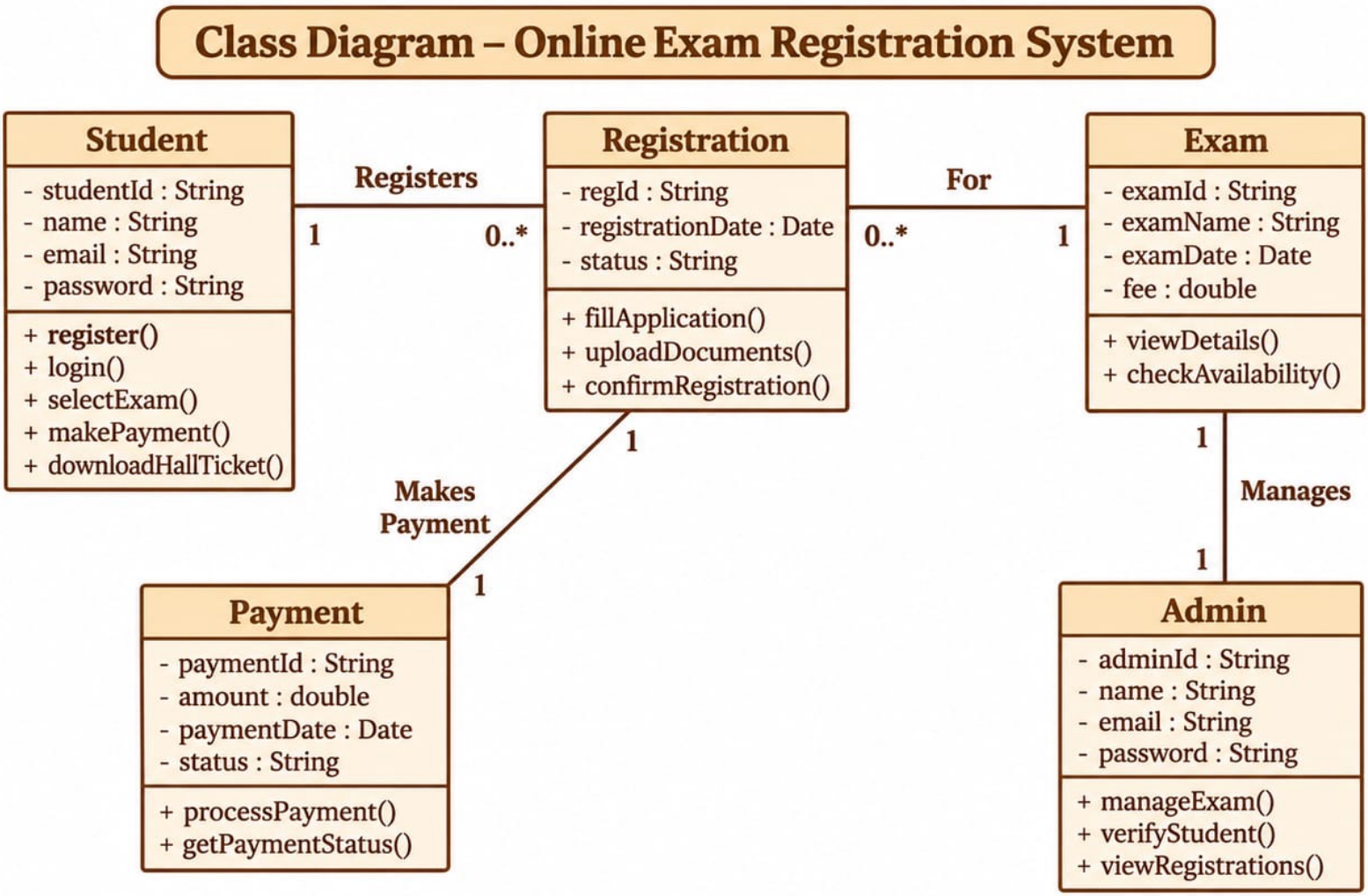
Therefore, an Online Exam Registration System is proposed. It will allow students to register online, enter their details, upload documents, pay fees, and check their registration status. The system will help administrators verify applications, manage records, and generate hall tickets automatically. This reduces paperwork, saves time, minimizes errors, and makes the registration process faster and more convenient.

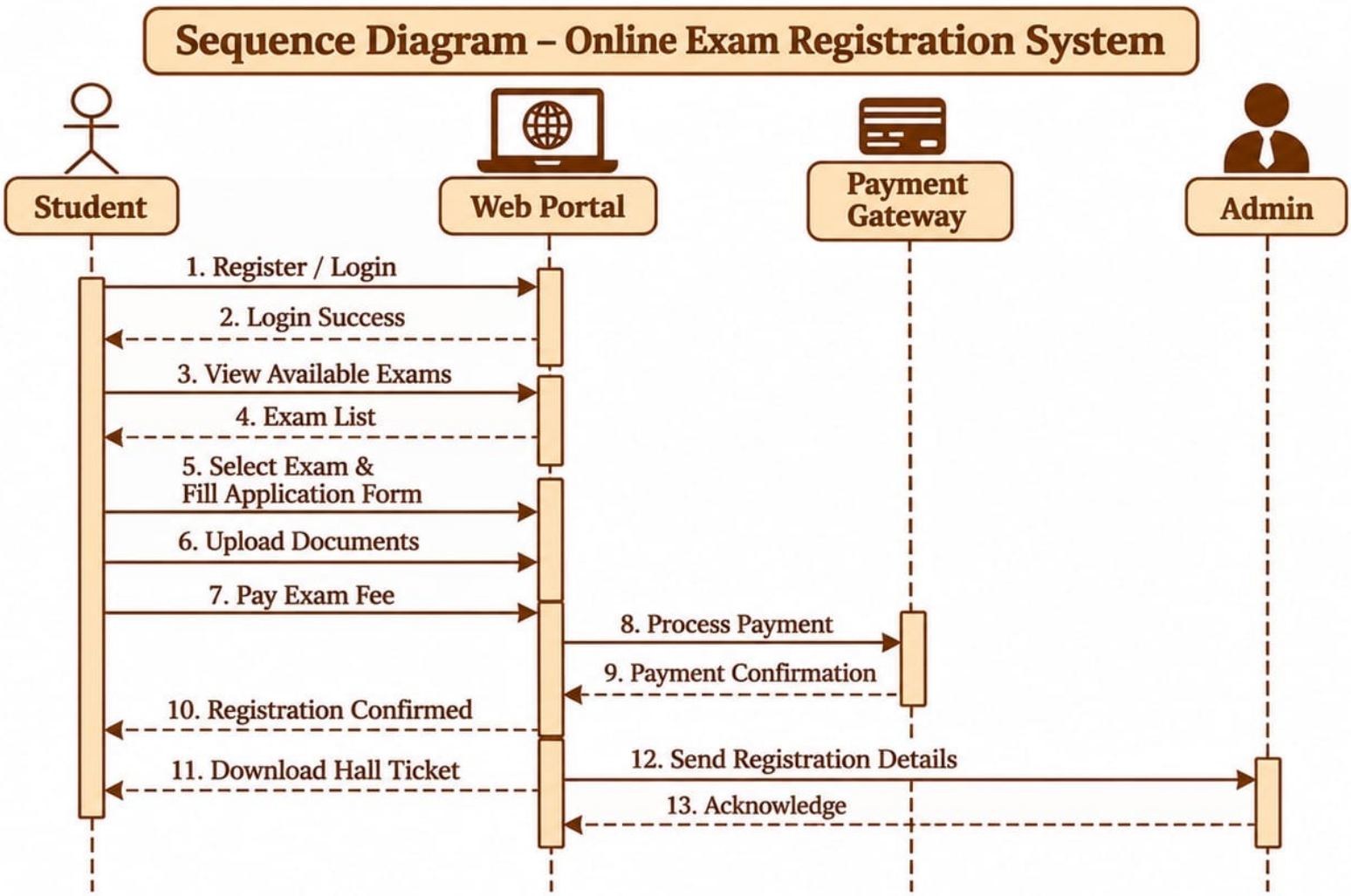

Timing Diagram – Online Exam Registration System
Time
Student
System
Database
Payment Gateway
t0
t1
t2
t3
t4
Login
Verify User
Process Registration
Store User Data
Initiate Payment
Save Registration Details
Process Payment
Payment Response
Show Success Message

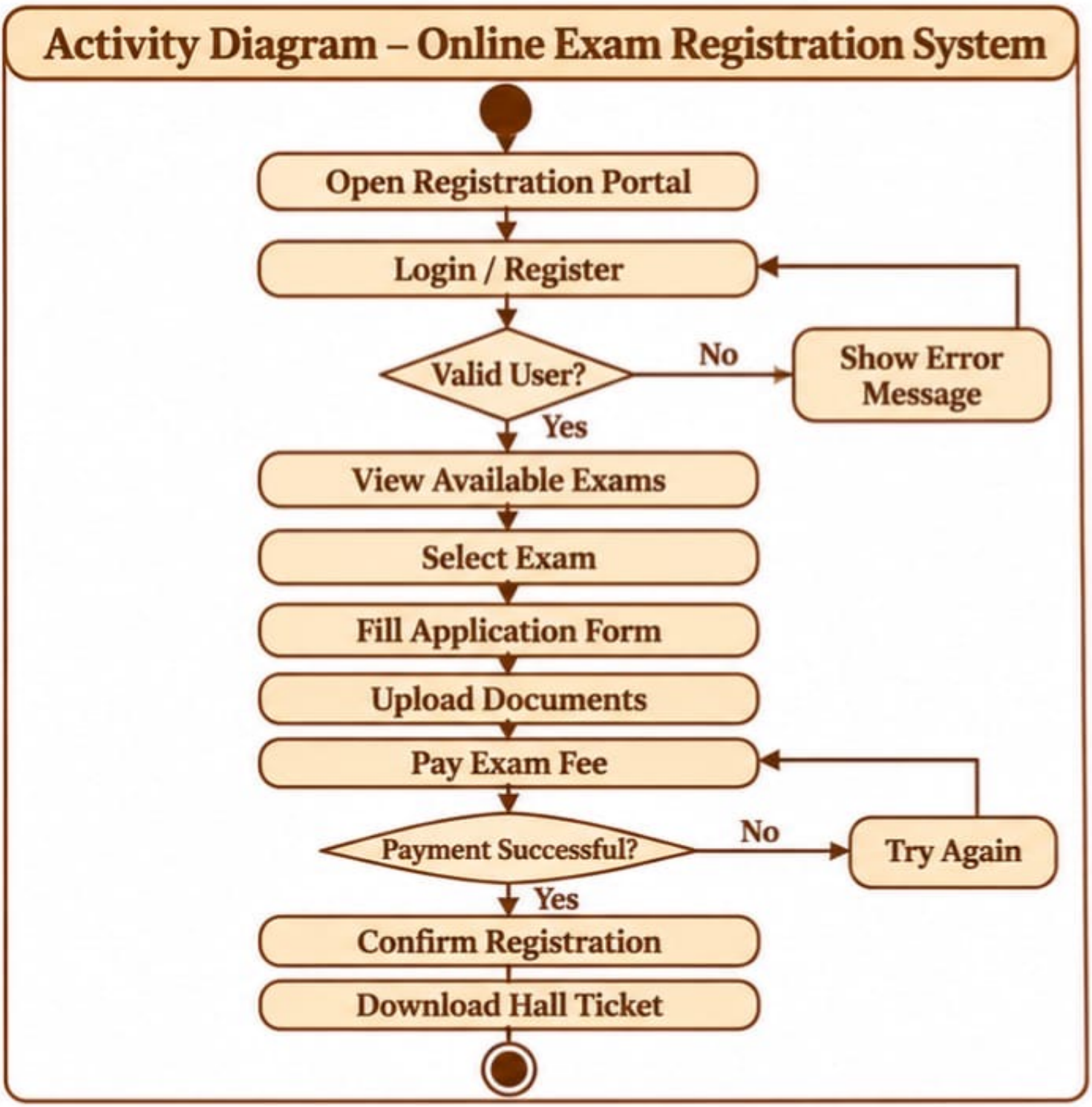

Object Diagram – Online Exam Registration System
: Student
studentId = S101
name = Priya
email = priya@mail.com
Registers
: Exam
examId = E101
name = JEE
date = 15-12-2026
Submits
: Application
appId = A1001
status = Pending
fee = 1000
Includes
: Document
docId = D201
type = ID Proof
fileName = id.pdf
For
Processed by
: Admin
adminId = A01
name = Sneha
Handles
: Payment
paymentId = P301
amount = 1000
status = Success

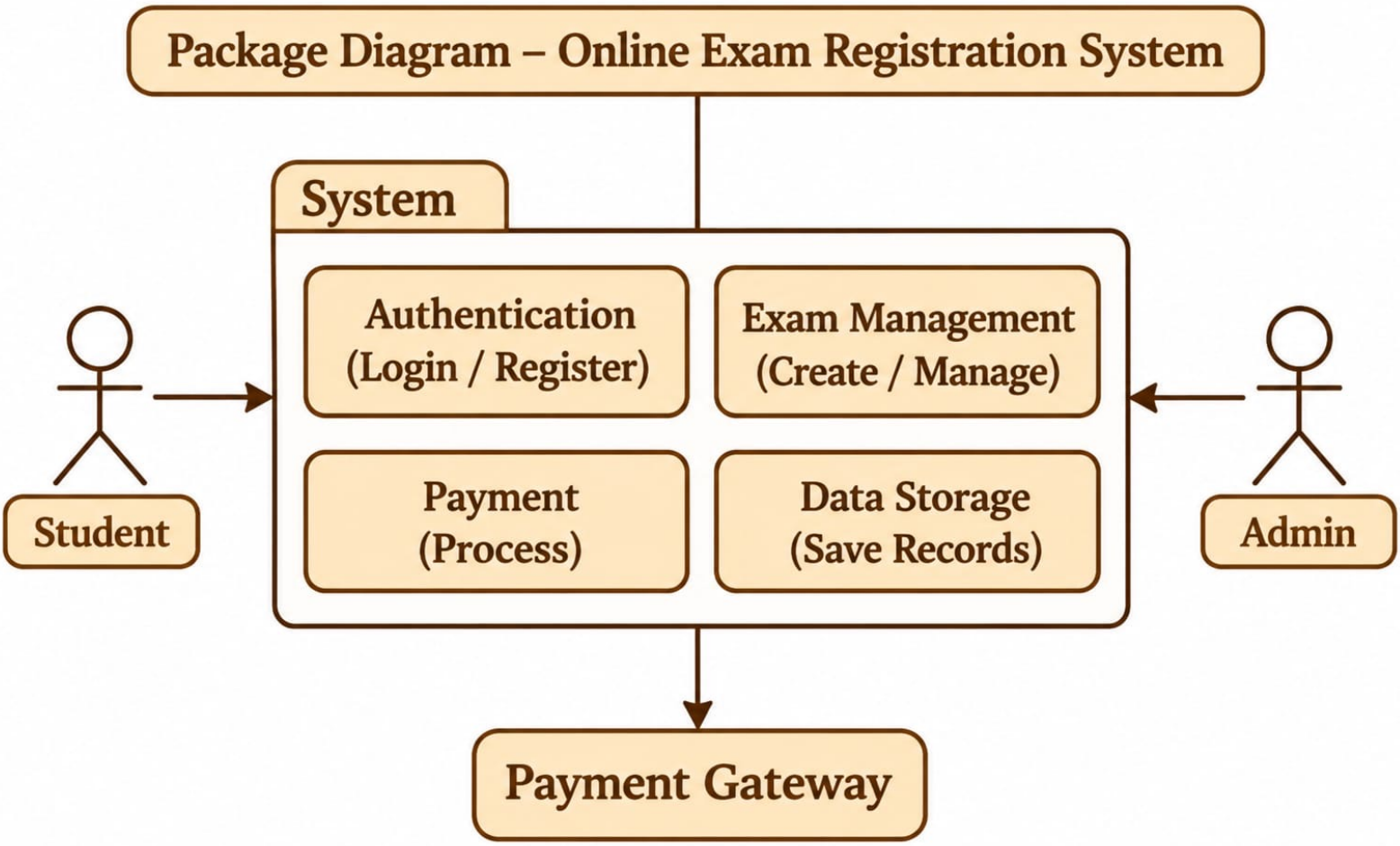
Component Diagram – Online Exam Registration System



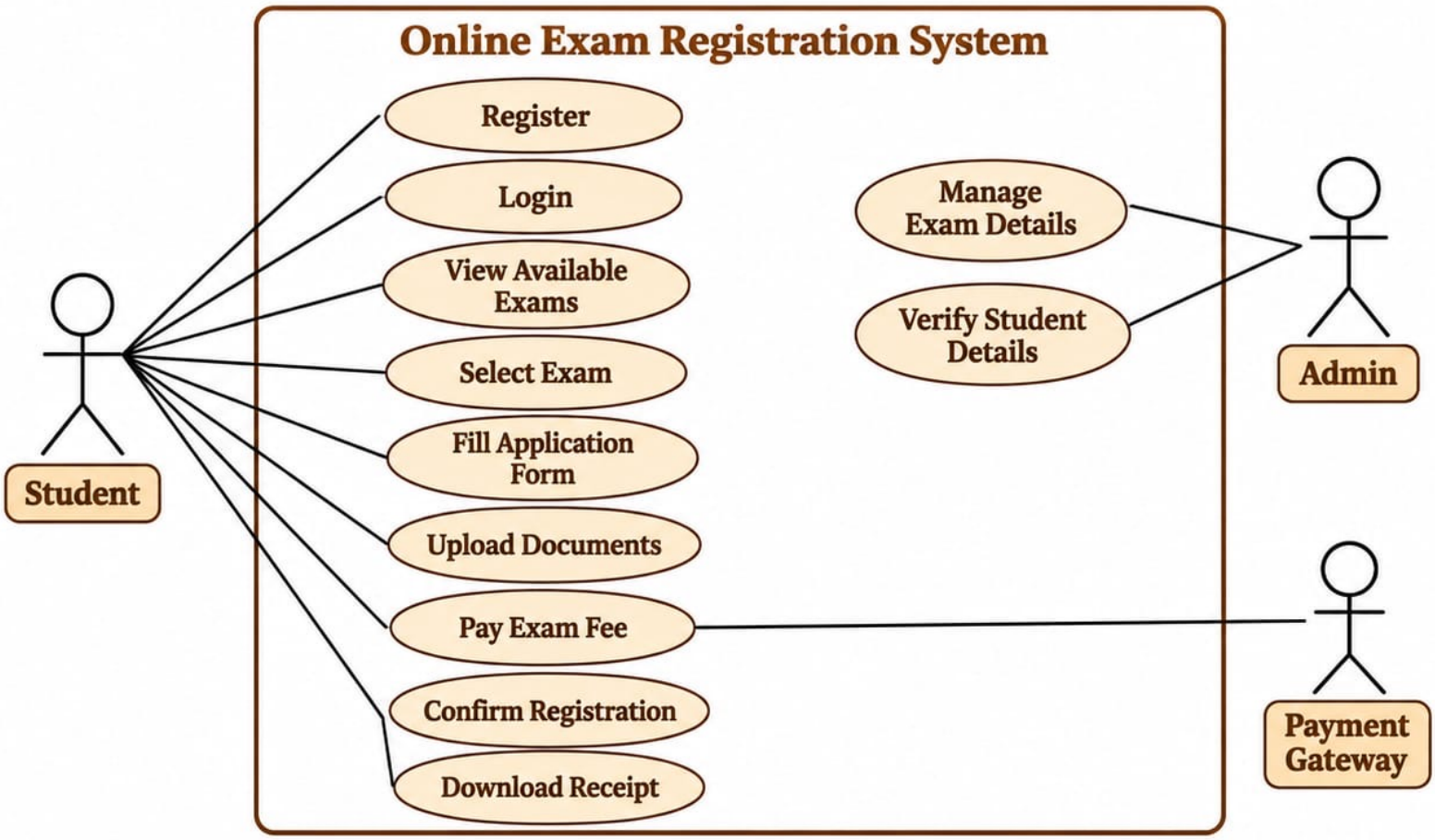

Class Diagram – Online Exam Registration System
Student
- studentId : String
- name : String
- email : String
- password : String
+ register()
+ login()
+ selectExam()
+ makePayment()
+ downloadHallTicket()
Registers
1
0..*
Registration
- regId : String
- registrationDate : Date
- status : String
+ fillApplication()
+ uploadDocuments()
+ confirmRegistration()
For
0..*
1
Exam
- examId : String
- examName : String
- examDate : Date
- fee : double
+ viewDetails()
+ checkAvailability()
1
Manages
1
Admin
- adminId : String
- name : String
- email : String
- password : String
+ manageExam()
+ verifyStudent()
+ viewRegistrations()
1
Makes Payment
1
Payment
- paymentId : String
- amount : double
- paymentDate : Date
- status : String
+ processPayment()
+ getPaymentStatus()


Sequence Diagram – Online Exam Registration System
Student
Web Portal
Payment Gateway
Admin
1. Register / Login
2. Login Success
3. View Available Exams
4. Exam List
5. Select Exam & Fill Application Form
6. Upload Documents
7. Pay Exam Fee
8. Process Payment
9. Payment Confirmation
10. Registration Confirmed
11. Download Hall Ticket
12. Send Registration Details
13. Acknowledge


Activity Diagram – Online Exam Registration System
Open Registration Portal
Login / Register
Valid User?
No
Show Error Message
Yes
View Available Exams
Select Exam
Fill Application Form
Upload Documents
Pay Exam Fee
Payment Successful?
No
Try Again
Yes
Confirm Registration
Download Hall Ticket


Package Diagram – Online Exam Registration System
System
Authentication
(Login / Register)
Exam Management
(Create / Manage)
Payment
(Process)
Data Storage
(Save Records)
Student
Admin
Payment Gateway

Use Case Diagram – Online Exam Registration System



TASK 2: Preparation of Software Requirement Specification (SRS) Document for Online Exam Registration System

Step 1: Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) document describes what the Online Exam Registration System should do and the constraints under which it operates.

Introduction

- **a) Project Name:** Online Exam Registration System
- **b) Purpose:** To provide an online platform for students to register for examinations, select subjects, pay examination fees, and download hall tickets.
- **c) Scope:** Students can register for exams through a website using their login credentials. Administrators can manage student details, exam schedules, and registration records.

1. Functional Requirements (What the system does)

1. User registration.
2. User login.
3. Student profile management.
4. Exam selection.
5. Subject selection.
6. Exam fee payment.
7. Registration confirmation.
8. Hall ticket generation.
9. Registration status checking.
10. Logout.

- **a) User Registration:** Allows new students to create an account by entering their name, email, phone number, and student ID.
- **b) User Authentication:** Verifies the student's username and password before allowing access.
- **c) Exam Selection:** Allows students to select available examinations according to their course and semester.
- **d) Subject Selection:** Allows students to choose the subjects for which they want to appear.
- **e) Fee Payment:** Allows students to pay examination fees through online payment methods.
- **f) Confirmation Generation:** Generates a registration confirmation after successful payment.
- **g) Hall Ticket Generation:** Generates a hall ticket containing student details, exam date, time, and examination centre.
- **h) Status Checking:** Allows students to check their registration and payment status.
- **i) Admin Management:** Allows administrators to add exams, update schedules, manage student details, and view registrations.
- **j) Logout:** Allows users to securely exit their accounts.

2. Detailed Functional Requirements

1. User Registration

- **Description:** Allows new students to create an account by entering personal and academic details.
- **Input:** Student Name, Student ID, Email, Phone Number, Password.
- **Output:** Successful registration message.

2. **User Login**

- **Description:** Allows registered students to log in using their email or student ID and password.
- **Input:** Username and Password.
- **Output:** Student dashboard.

3. **Student Profile Management**

- **Description:** Allows students to view and update their personal and academic information.
- **Input:** Name, Contact Number, Address, Course Details.
- **Output:** Updated student profile.

4. **Exam Selection**

- **Description:** Displays available examinations according to the student's course and semester.
- **Input:** Course, Semester, Exam Name.
- **Output:** Available examination details.

5. **Subject Selection**

- **Description:** Allows students to select the subjects for the examination.
- **Input:** Subject Names and Subject Codes.
- **Output:** Selected subject list.

6. **Exam Fee Payment**

- **Description:** Allows students to pay examination fees using online payment methods such as UPI, debit card, or net banking.
- **Input:** Student Details and Payment Information.
- **Output:** Payment confirmation or failure message.

7. **Registration Confirmation**

- **Description:** Generates a confirmation message after successful examination registration and fee payment.
- **Input:** Successful registration and payment details.
- **Output:** Registration Number and Payment Receipt.

8. **Hall Ticket Generation**

- **Description:** Generates a hall ticket containing student details and examination information.
- **Input:** Confirmed registration details.
- **Output:** Downloadable Hall Ticket.

9. **Registration Status**

- **Description:** Allows students to check whether their examination registration is completed, pending, or rejected.
- **Input:** Student ID or Registration Number.
- **Output:** Current Registration Status.

10. **Logout**

- **Description:** Terminates the active session and securely logs the user out of the system.
- **Input:** Logout request.
- **Output:** Login Page.

3. Non-Functional Requirements (System Attributes)

- **a) Security:** Protects student information using passwords, authentication, and data encryption.
- **b) Performance:** The system should respond to user requests within 3 seconds under normal conditions.

- **c) Availability:** The system should be available 24 hours a day, 7 days a week.
- **d) Usability:** Provides a simple and user-friendly interface for easy registration.
- **e) Reliability:** Ensures accurate registration records and payment details.
- **f) Scalability:** Supports a large number of students registering at the same time.

4. Detailed Non-Functional Requirements

1. Security Through User Authentication

- **Password Protection:** Passwords must be stored securely using encryption or hashing.
- **Secure Login:** Only authorized users can access student accounts.
- **Session Management:** Automatically logs out users after a period of inactivity.

2. Fast Response Time

- **Low Latency:** Registration and other operations should complete within 3 seconds under normal conditions.
- **Quick Loading:** Web pages should load quickly without unnecessary delays.

3. 24×7 Availability

- **High Availability:** The system should remain available throughout the day.
- **Backup:** Regular backups should be maintained to prevent data loss.

4. Reliability and Accuracy

- **Data Accuracy:** Student details, subject selection, and payment records must be accurate.
- **Error Handling:** The system should display proper error messages when registration or payment fails.
- **Data Recovery:** The system should recover important information in case of system failure.

5. User-Friendly Interface

- **Easy Navigation:** Provides simple menus and clear instructions for students.
- **Accessibility:** Supports different devices such as mobiles, tablets, and computers.

6. Data Confidentiality

- **Data Encryption:** Protects student information during transmission.
- **Data Privacy:** Student personal details and payment information must remain confidential.
- **Access Control:** Only authorized administrators can access and modify student records.

Table 1: Hardware Requirements

Hardware Components	Description and Specification
Server	Stores the application and student database.
Computer / Laptop	Used by students and administrators to access the system.
RAM	Minimum 4 GB RAM for client devices.
Hard Disk	Provides storage for application files and database records.
Network Device	Router or modem for internet connectivity.

Table 2: Software Requirements

Software Components	Description and Specification
Operating System	Windows or Linux.
Frontend	HTML, CSS, and JavaScript.
Backend	Java, Python, or PHP.
Database	MySQL for storing student, examination, and payment records.
Web Browser	Google Chrome, Mozilla Firefox, or Microsoft Edge.
Web Server	Apache or Tomcat.